

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities



429
ppm

A Data-Driven Overview | 1750 – 2024

Source: Global Carbon Budget 2025 · NASA · Our World in Data

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

February 2026 — Mauna Loa Observatory

+53% from pre-industrial ~280 ppm

Acceleration of Atmospheric CO₂

●	Pre-Industrial (before 1750)	~280 ppm	Natural baseline
●	1958 — Keeling Curve begins	~315 ppm	+12.5% from baseline
●	2000	~370 ppm	+32% from baseline
●	2020	~414 ppm	+48% from baseline
●	Feb 2026	429 ppm	+53% from baseline

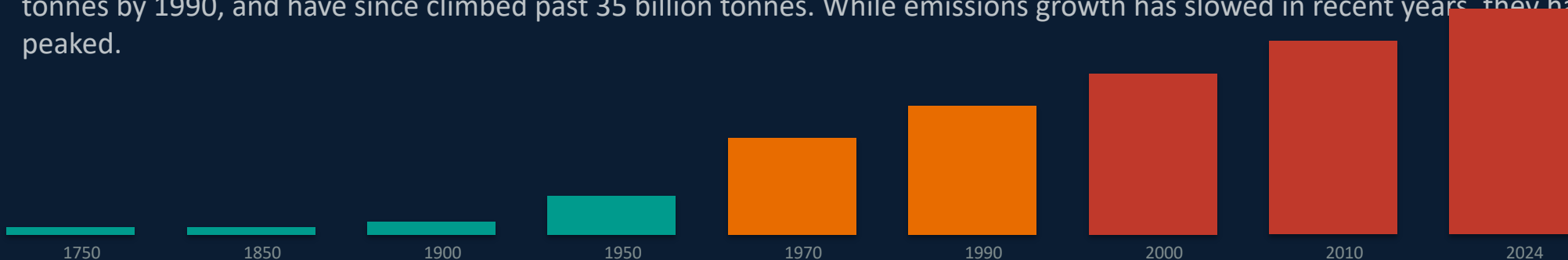
The current rate of increase is at least 1.5× greater than the natural increase observed at the end of the last ice age — a change that took thousands of years then but has occurred in roughly two centuries now.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Bt to 35+ Bt



Emissions Trajectory (1750–2024)

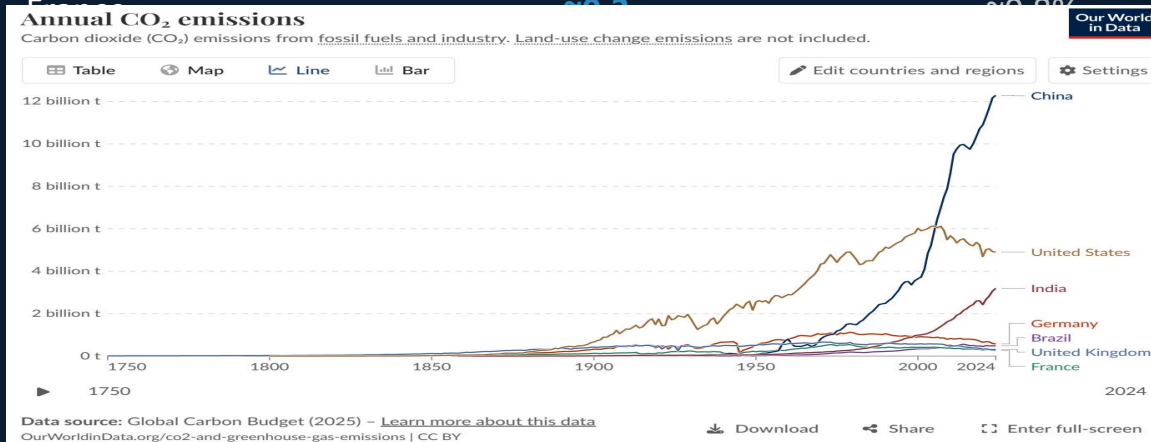
Before the Industrial Revolution, CO₂ emissions were negligible. Growth remained relatively slow until the mid-20th century, when rapid industrialisation drove a steep increase. From roughly 6 billion tonnes in 1950, emissions nearly quadrupled to over 20 billion tonnes by 1990, and have since climbed past 35 billion tonnes. While emissions growth has slowed in recent years, they have not yet peaked.



Billion tonnes CO₂/year (approximate)

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country / Region	Annual CO ₂ (Bt)	Share of Global	Recent Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.2	~0.5%	Declining

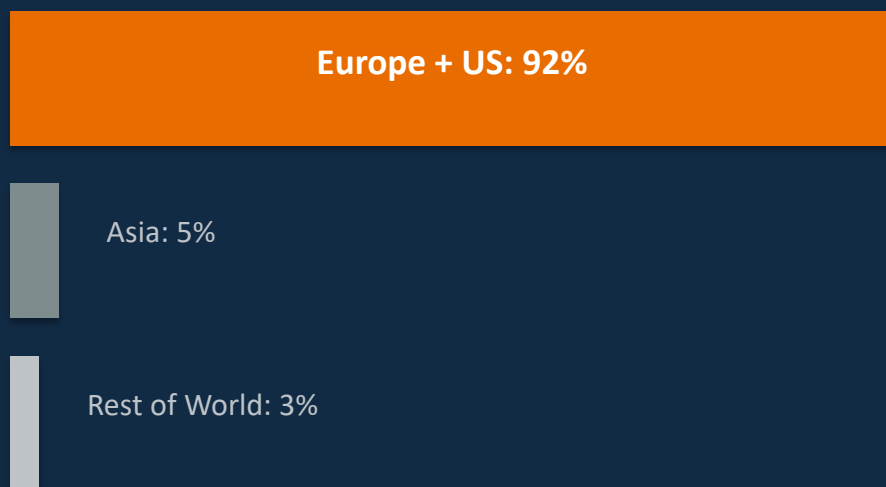


Key Insight

China's steep rise post-2000 has made it the dominant emitter. The US peaked near 6 Bt around 2005 and has since declined to ~5 Bt. India's rapid rise positions it as the third-largest emitter.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

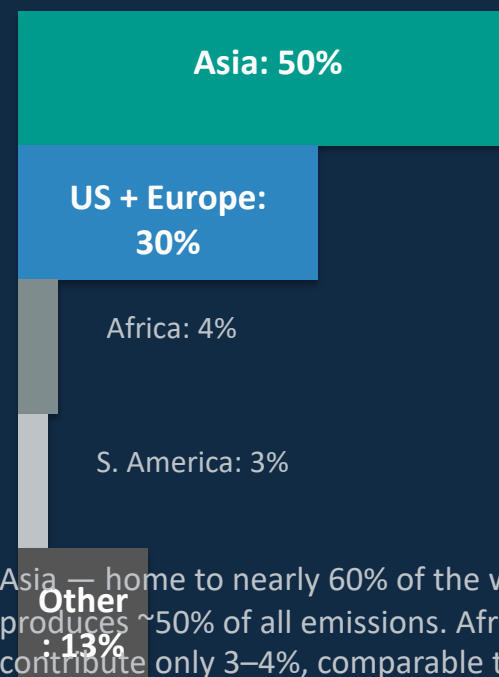
1900 — Europe & US Dominate



In 1900, Europe and the US accounted for more than 90% of global emissions. By 1950, they still represented over 85%.

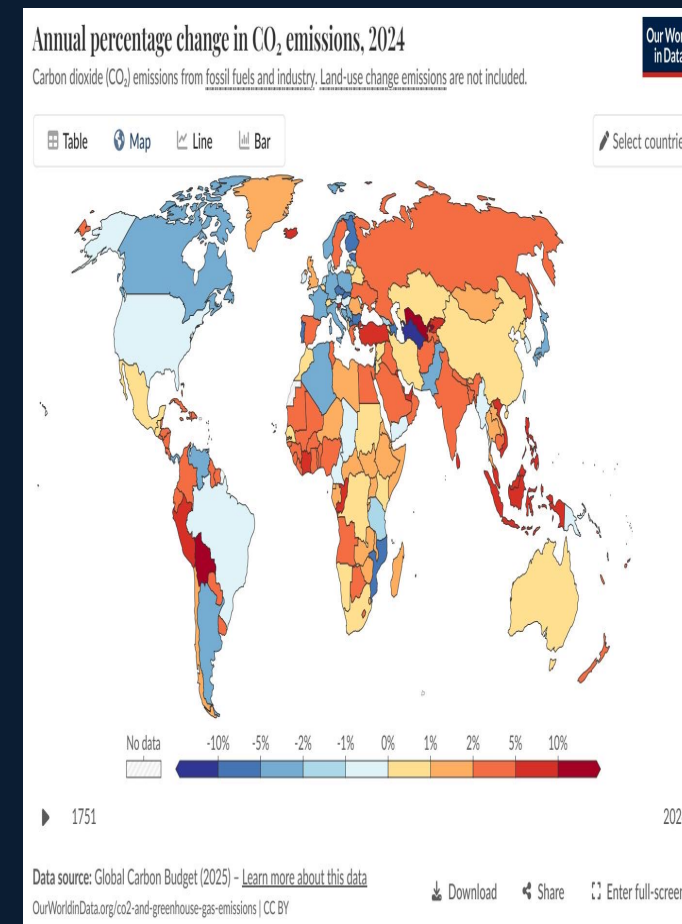
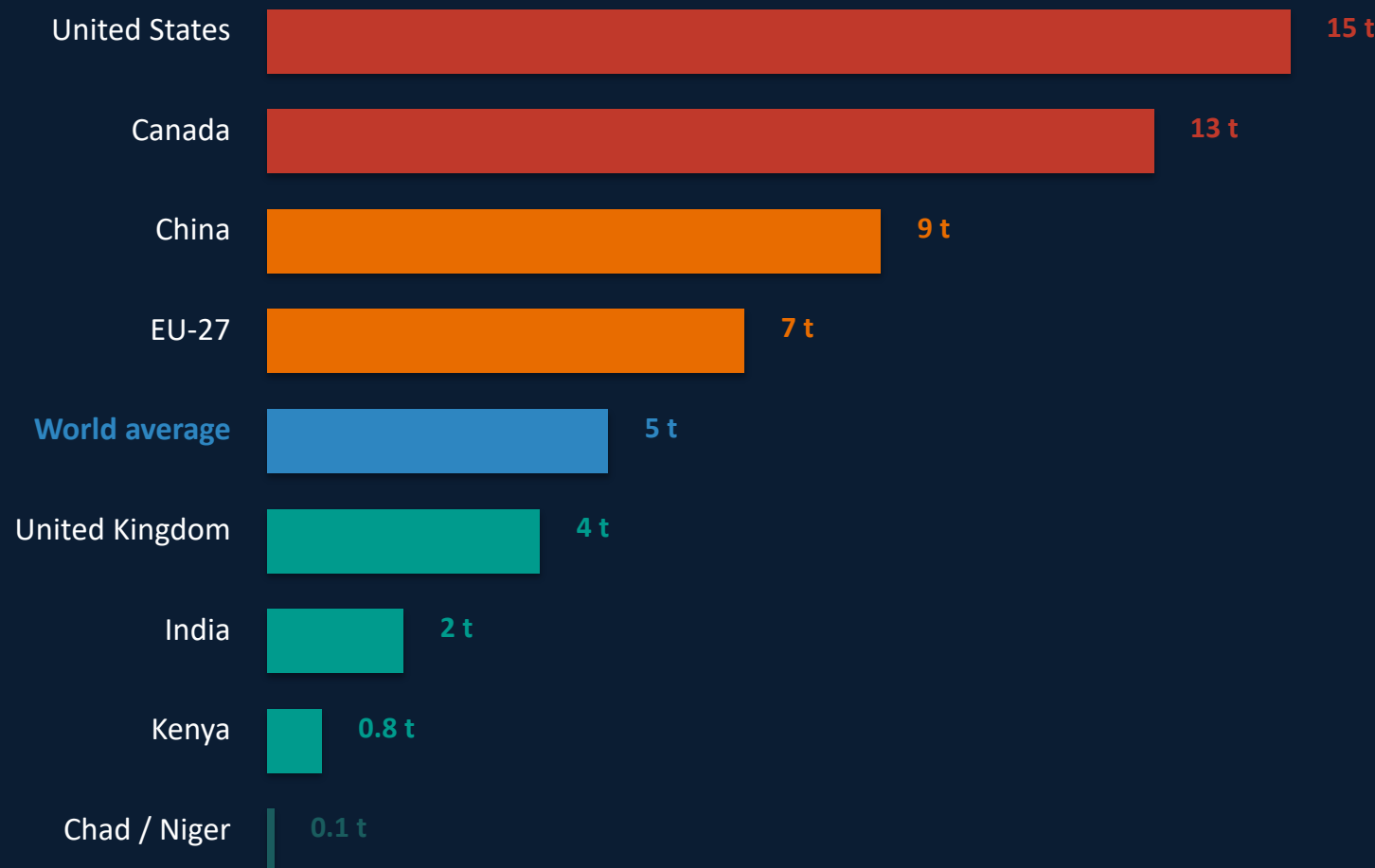


2024 — Asia Ascendant



Asia — home to nearly 60% of the world's population — now produces ~50% of all emissions. Africa and South America each contribute only 3–4%, comparable to international aviation and shipping.

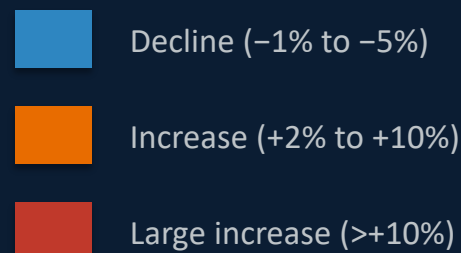
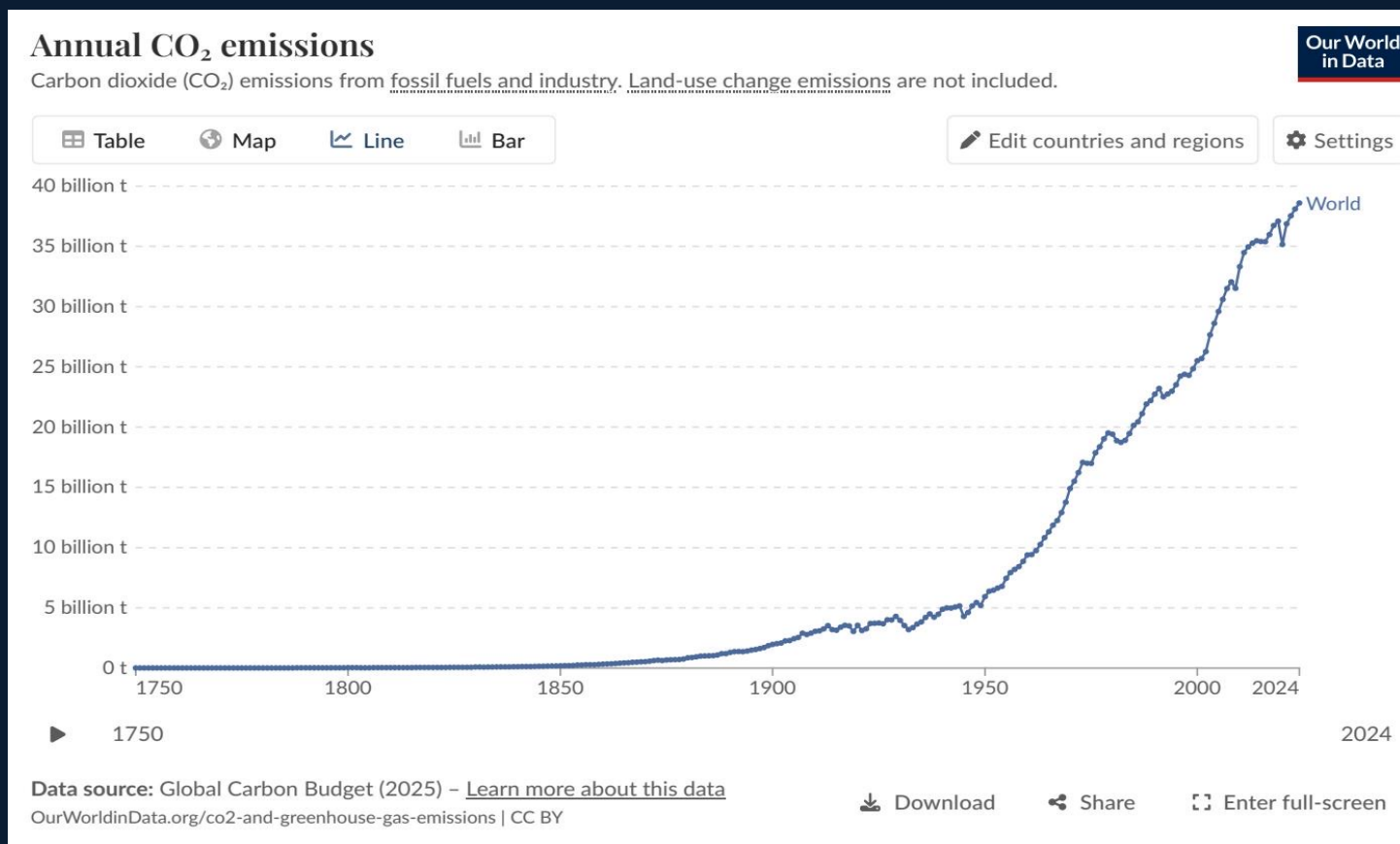
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



The average American emits in under two days what the average person in Chad or Niger emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

2024 Annual Change in CO₂ Emissions



Regional Patterns

- US & much of Europe: declining (-1% to -5%)
- Russia: increasing (+2% to +10%)
- Southeast Asia: increasing
- Parts of Africa: mixed to increasing
- Some S. American nations: >+10%

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their energy sources. Nuclear and renewables displace fossil fuels, lowering per capita CO₂ significantly.

Country	Nuclear Share	Renewables Share	Fossil Fuel Share	Per Capita CO ₂ (t)
Lower Per Capita Emitters				
France	~70%	~15%	~15%	~4
United Kingdom	~20%	~40%	~40%	~4
Higher Per Capita Emitters				
Germany	~5%	~45%	~50%	~8
United States	~0%	~10%	~90%	~15

Policy & Energy Mix Matter More Than Prosperity Alone

France's heavy reliance on nuclear power gives it per capita emissions roughly half those of Germany, despite similar GDP per capita. The UK's shift toward renewables and away from coal has similarly reduced its per capita footprint. These differences demonstrate that energy policy choices — not just wealth — determine per capita CO₂ outcomes.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- 1 Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
- 2 China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.
- 3 The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today.
- 4 Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US ~15 t/person) from the lowest (Chad ~0.1 t/person).
- 5 Policy and energy mix choices — not just prosperity — determine per capita outcomes. France's nuclear-heavy mix yields half the per capita emissions of Germany.